

# A versioned, leakage-audited USGS *Did You Feel It?* benchmark for machine-learning evaluation of severe felt intensity

Akshay Sharma<sup>1\*</sup> and Lalji Prasad<sup>1</sup>

<sup>1\*</sup>Department of Computer Science and Engineering, SAGE University, Indore, 452020, Madhya Pradesh, India.

\*Corresponding author(s). E-mail(s): [akssha74@gmail.com](mailto:akssha74@gmail.com);  
Contributing authors: [lalji.research@gmail.com](mailto:lalji.research@gmail.com);

## Abstract

Reliable comparison of models that estimate earthquake felt intensity requires a stable reference dataset and evaluation rules that prevent related events from crossing data splits. We introduce version 1 of a single-source USGS “Did You Feel It?” (DYFI) benchmark for retrospective prediction of severe felt intensity (event-maximum community decimal intensity  $\geq 6$ ) from earthquake source metadata. The resource combines a frozen dataset, a six-category leakage audit, internally specified space-time sequence grouping, outcome-independent analysis roles, endpoint sensitivity analysis, fixed baselines, calibration-aware scoring, Brier-score sequence-cluster bootstrap uncertainty, and a protected temporal evaluation. The frozen  $M \geq 5$  snapshot contains 7,520 records, 2,362 eligible events, and a 430-event temporal holdout. The cross-validation comparison yielded small differences favoring sequence grouping: its Brier score was 0.006 lower and its area under the receiver-operating-characteristic curve (AUROC) was 0.012 higher than under random folds. Because the executed analysis retained only point estimates for this contrast, we interpret it descriptively. All learned baselines outperformed the no-skill reference, but the paired 95% interval between the two lowest-Brier models included zero; therefore, we identify no winning model. The benchmark provides a versioned, leakage-audited reference for future methods, and every reported value can be reconstructed exactly from the frozen analysis files.

**Keywords:** Did You Feel It?, macroseismology, machine learning, benchmark resource, sequence-grouped cross-validation, reproducibility

## Article Highlights

- A frozen USGS felt-report resource enables reproducible comparison of intensity models.
- Events linked by the fixed space-time rule remain together during evaluation.
- Split differences were small, and no model was declared the winner.

## Acknowledgements

We thank the USGS DYFI and ANSS Comprehensive Catalog programs.

# 1 Introduction

The U.S. Geological Survey “Did You Feel It?” (DYFI) system converts crowd-sourced felt reports into a numeric community-decimal intensity (CDI) for each earthquake, providing a dense, decades-long record of observed shaking that complements sparse instrumental networks (Wald et al. 1999; Atkinson and Wald 2007; Wald et al. 2012; Quitoriano and Wald 2020). Because the felt-intensity outcome of an event is only realised after responses accumulate, a natural retrospective question is how well the *severe* outcome of an event—an event-maximum intensity in the damaging range ( $\text{maxCDI} \geq 6$ )—can be anticipated from outcome-blind source metadata in the frozen catalogue snapshot (magnitude, depth, location, origin time). These are preferred catalogue values at the snapshot instant, not guaranteed first-alert or real-time values. To protect the temporal hold-out, however, the primary models use only magnitude and depth; location and origin time appear only in restricted sensitivity analyses. This defines a descriptive, retrospective target quantity, useful for characterising how much of the severe-shaking signal is carried by catalogue parameters alone, and it is distinct from any real-time, operational, or causal claim.

A recurring methodological problem undermines the machine-learning literature on this and adjacent tasks. Models are trained and evaluated on datasets assembled ad hoc from the earthquake catalogue and split either at the record level or by a single random train/test partition. Random splitting ignores the dependence structure of seismicity: aftershocks of a mainshock share source parameters and overlapping felt fields, so a random split can place near-duplicate events on both sides of the partition and inflate apparent skill—a textbook form of data leakage (Kaufman et al. 2012) that grouped or blocked cross-validation is designed to remove (Roberts et al. 2017). Recent macroseismic-intensity studies illustrate the pattern: models are fit on single-system instrumental or DYFI data under random or stratified partitions, with no released, leakage-audited, event-grouped evaluation resource that a later user could reuse without reconstructing a contaminated split (Patelli et al. 2026; Mori and Naso 2025). The consequence is that reported numbers are hard to reproduce and easy to over-read.

## 1.1 Contributions

We address the reproducibility and leakage gap by introducing a versioned benchmark resource centered on a reusable, leakage-audited evaluation contract. Concretely, the resource provides:

1. **A single-source, version-frozen DYFI benchmark** for the exact task “outcome-blind source metadata  $\rightarrow$  later event-maximum  $\text{maxCDI} \geq 6$ ”, frozen to one dated access instant so that any later catalogue revision defines a new version rather than a silent overwrite (Section 3).
2. **An executable six-category leakage audit** that rejects direct-label, post-outcome-ascertainment, identifier-proxy, and downstream-product features, and rejects cross-role sequence straddles and temporal-cutoff violations, accepting *only* the exact source-only feature set; it is exercised by 110 unit tests and rejects every deliberately contaminated test case (Section 4.1).
3. **Internally specified, outcome-independent roles** built on order-stable space-time sequence grouping and a temporal external holdout, so that split membership never uses the label (Section 4.2).
4. **A fixed, non-selective baseline suite** with calibration-aware proper scoring, Brier-score sequence-cluster bootstrap uncertainty, explicit handling of ties and unavailable calibration estimates, and single-use authorization of the decision-bearing protected evaluation (Sections 4.4–4.5).
5. **A quantified split-sensitivity diagnostic** (naive random versus sequence-grouped splitting) reported as a descriptive point estimate, together with a fully reproducible release bundle and machine-readable verification record (Section 5).

The learned models establish reference performance for the benchmark. Their comparisons do not identify a winner, and the study makes no cross-system, real-time, lead-time, operational, or causal claim.

## 1.2 Significance

The resource gives later studies a fixed, auditable contract instead of another bespoke split. By combining a version-frozen macroseismic corpus,

an executable leakage audit, sequence-grouped roles, reference baselines, and exact reconstruction, it turns citizen-reported felt intensity into a reusable seismological testbed rather than proposing another standalone prediction model.

## 2 Related work and novelty

### *DYFI and macroseismic intensity.*

The DYFI system and its CDI construction, outlier handling, and post-event revision behaviour are documented in the foundational and review literature (Wald et al. 1999; Atkinson and Wald 2007; Wald et al. 2012; Quitoriano and Wald 2020); these establish that CDI is an after-the-event, response-driven quantity subject to ascertainment effects, which is precisely why our protocol uses response count only to determine eligibility and forbids it as a predictive feature (Section 3). Rösler et al. (2022) further show specifically for DYFI that interpreted products can change and should be preserved with timestamps for reproducible analysis. The dated, hashed snapshot released here implements that preservation principle. We reuse these provenance facts rather than re-deriving them.

### *Domain benchmark resources.*

The INGe resource links Italian macroseismic intensities to colocated strong-motion records for benchmarking intensity-ground-motion relationships (Oliveti et al. 2022). It is a domain-near resource precedent, but differs from the present event-level USGS DYFI classification task, source-only feature boundary, leakage audit, and protected temporal evaluation. The 2026 USGS Western North America resource contains about 696,000 site-level macroseismic observations from about 5,600 earthquakes and is predominantly DYFI (Fathian Sabet et al. 2026). It is a larger observational resource, but does not define this event-maximum binary task, executable leakage audit, or frozen temporal evaluation. Stockman et al. (2026) provide the nearest broader leakage-aware seismological benchmark precedent: EarthquakeNPP standardizes public catalogues, repairs an improper earthquake-forecasting split, compares neural point processes with ETAS, and finds no consistent neural-model advantage. Its target is occurrence forecasting with chronological

likelihood/CSEP evaluation, rather than source metadata predicting a later citizen-reported felt-intensity endpoint; the resources are therefore methodologically adjacent but task-distinct. An adjacent event-level binary study used post-event LastQuake felt reports to classify high- versus low-impact earthquakes and evaluated later 2022 events (Lilienkamp et al. 2023). Its crowdsourced predictors, impact endpoint, and rapid-response objective differ from the source-only DYFI benchmark and reusable evaluation contract developed here.

### *Macroseismic databases and citizen observations.*

Event-specific analysis in the 2019 Athens earthquake combined online felt reports with ground motion and source modelling (Kouskouna et al. 2021). Crowdsourced macroseismic observations have also been used to characterize geological settings and spatial variations in shaking response (Saraò et al. 2023). Curated macroseismic resources such as the Belgian traditional macroseismic database and MACROSISSDATA preserve historical intensity observations for transparent reuse and later analysis (Neefs et al. 2025; Sira et al. 2025). The present work extends this seismological resource tradition from data preservation to a versioned, executable contract for model evaluation.

### *Machine learning on felt/macroseismic intensity.*

Recent studies predict macroseismic intensity with random forests and gradient boosting from instrumental or catalogue features (Patelli et al. 2026; Mori and Naso 2025). Machine learning has also been applied to earthquake alert-level classification using intensity-related features and to waveform detection with openly released seismic datasets (Özüpak 2026; Ertuncay et al. 2025). Those tasks use post-event intensity descriptors or waveforms, rather than the source-only feature boundary studied here. Each felt-intensity model application is evaluated under a random or stratified partition and does not release a reusable, event-grouped, leakage-audited benchmark; Mori and Naso (2025) moreover predict from instrumental ground-motion parameters rather than outcome-blind catalogue metadata, and Patelli

et al. (2026) evaluate a single-system train/test split on seismometric inputs. Their concrete drawback for our purpose is the evaluation contract: a random or stratified split does not remove within-sequence near-duplicate contamination, and none ships an executable leakage audit or a version-frozen resource that a third party can rerun. Our protocol addresses that specific drawback with internally specified sequence-grouped roles and an executable six-category audit.

#### ***DYFI-based intensity models and concurrent leakage-aware work.***

Closest to our exact data source, Karthik Kumar et al. (2025) predict DYFI community intensity from the same catalogue metadata (magnitude, depth, location) with a random-forest regressor on Indian events under a single naive train/test split (reported  $R^2 = 0.829$ ); our resource does not fit a competing regressor but instead supplies a versioned, digitally fingerprinted, event- and sequence-grouped benchmark with an executable leakage audit and a protected decision-bearing evaluation authorized once, in place of the naive split that setup uses. Concurrently, Li et al. (2025) release a DYFI-grounded benchmark that simulates Modified Mercalli Intensity (MMI) for two earthquakes (2014 Napa and 2019 Ridgecrest) with large language models and includes an explicit location-identifier leakage test; our resource differs in task and construction—an outcome-blind source-metadata→later-severe binary classification over a decade-scale  $M \geq 5$  corpus with internally specified sequence-grouped roles, rather than multimodal per-zip-code MMI simulation for two events—so the two are complementary rather than duplicative.

#### ***Cross-validation for dependent data.***

That grouped or blocked cross-validation is the correct evaluation strategy for temporally, spatially, or hierarchically structured data is established practice (Roberts et al. 2017; Kaufman et al. 2012); we do not claim this as novel.

#### ***Novelty delta.***

Conceding the above, the delta this resource supplies that prior work does not is the *released, versioned, hashed, single-source DYFI benchmark*

itself: an executable six-category audit covering forbidden fields, proxy variables, and split contamination; an internally specified, sequence-grouped role assignment with a temporal external holdout, a documented severe endpoint with threshold and threshold sensitivity, a fixed non-selective baseline suite with a no-skill comparator, a calibration/uncertainty/tie contract, an exact-reproduction verification, and a quantified naive-versus-grouped split diagnostic. A current-year search located no released resource occupying this exact combination; novelty is stated as a concrete delta to located work, with no first/only superlative asserted.

### **3 Data, provenance, and outcome-boundary disclosure**

#### ***Source and single-instant pinning.***

The corpus is derived entirely from USGS DYFI and its parent ANSS Comprehensive Catalog (ComCat), retrieved through the International Federation of Digital Seismograph Networks (FDSN) event web service (Wald and Dewey 2005; U.S. Geological Survey, Earthquake Hazards Program 2017). The acquisition envelope is outcome-independent:  $M \geq 5$ , product type `dyfi`, event type `earthquake`, starttime 2015-01-01 to endtime 2025-01-01. FDSN defines the end bound as inclusive; no frozen record occurs at that boundary, so the realized snapshot spans 2015–2024. Results are ordered by time. The preliminary count request and the final data request use this same envelope. Because the catalogue evolves, the downloaded data were frozen at one access instant (2026-09-01T06:33:30Z) and assigned a SHA-256 digital fingerprint. A preliminary count changed from 7,575 to 7,576 during the acquisition session, documenting one live-catalogue revision. The final downloaded snapshot contained 7,520 DYFI records, and a count requested 50 seconds later in the same acquisition session also returned 7,520. Thus, the one-record change belongs only to the earlier preliminary checks; it is not a discrepancy in the frozen analysis snapshot. Likewise, the 56-record difference between the last preliminary count and the final 7,520-record snapshot is not an analysis exclusion: the preliminary count had no paired payload, so its membership difference cannot be decomposed and is retained

only as catalogue-version provenance. The near-contemporaneous final query and count agree exactly. The digital fingerprint of the downloaded bytes, rather than a preliminary count, defines this resource version. The frozen snapshot contains 7,520 DYFI records (Table 1, Fig. 1).

### ***Protocol chronology.***

The file records an internal 2026-09-01 freeze but was first publicly timestamped on 2026-09-19. We therefore describe it as an internally frozen protocol, not an externally registered preregistration.

### ***Retrospective eligibility and post-report ascertainment.***

DYFI intensity is an after-the-event quantity whose reliability grows with the number of felt reports (Quitoriano and Wald 2020). To keep the label meaningful we apply a data-informed eligibility filter of at least ten responses per event. The response count is used *only* for eligibility and is explicitly forbidden as a predictive feature, because it is a post-outcome ascertainment proxy; this forbiddance is enforced by the leakage audit (family F2, Section 4.1). Eligibility yields 2,362 events with severe prevalence 0.298 (703 severe events). Of these, 2,360 events are assigned to the internally specified analysis roles (Section 4.2) and contain 702 severe events; the remaining two are temporal-straddle exclusions, one of which is severe. That single severe straddle event accounts for the difference between the eligible cohort (703 severe) and the analysis-assigned set (702 severe), so eligibility is not the same as analysis-role assignment (Table 1).

### ***Endpoint.***

The severe label is severe =  $1[\text{event-maximum maxCDI} \geq 6.0]$  (intensity VI+). The primary threshold is 6.0, with reported sensitivity at {5.5, 6.0, 6.5} and a near-threshold band of  $\pm 0.5$  (Section 5.2).

### ***Outcome-boundary disclosure and held-out semantics.***

Only corpus-level outcome-blind counts and inherited aggregate feasibility statistics were exposed before construction; no per-event label informed the endpoint, split, or tuning choices, and role

membership is assigned by an internally specified, outcome-independent rule that never uses the label. We therefore describe the external holdout as *held out from model development with aggregate feasibility previously known*, and we deliberately do *not* call it “untouched” or “confirmatory.” Two event identifiers were prelisted for quarantine because their aggregate counts had been seen during source verification; neither identifier occurs in the frozen snapshot, so the rule is non-operative in version 1 and excludes no record. The complete list of protocol decisions is reported in Section 5.7. No respondent-level personally identifying information is redistributed; only event-level source metadata, the severe label, the eligibility-only response count, identifiers, and role assignments are released. Snapshot-level provenance is supplied by the activated source-locator manifest and source payload SHA-256; version 1 does not claim per-event product-version hashes. The release bundle follows datasheet-for-datasets documentation (Gebru et al. 2021) and findable, accessible, interoperable, and reusable (FAIR) data principles (Wilkinson et al. 2016).

## **4 Benchmark design**

### **4.1 Feature sets and the six-category leakage audit**

The primary *source-only* feature set contains magnitude and depth. It is the only feature set evaluated on the temporal holdout. An *expanded-metadata* sensitivity analysis (baseline B3) additionally uses latitude, longitude, and event year, so it is not evaluated where those variables would disclose the holdout axis. The audit accepts only an approved feature set with the expected fields in the expected order; any extra or unknown field is rejected.

The executable audit enforces six leakage families. Field-level: F1 direct label or its constituents; F2 post-outcome ascertainment or revision (including the response count, eligibility-only); F3 identifier or source proxy; F4 downstream-product proxy. Split-level: F5 sequence cross-role straddle; F6 temporal-cutoff violation. An axis-proxy check additionally rejects origin year under a temporal holdout and, when a geographic holdout is configured, latitude/longitude/region. The latter audit branch is provided for reuse but was not



executed as an external evaluation in this study. This enforces the B3 restriction. The audit is useful independently of model performance: it must reject every deliberately contaminated test case and accept only the approved clean feature set. It is exercised by 110 unit tests across four suites, all passing, and accepted the source-only feature set in the frozen snapshot.

## 4.2 Sequence grouping and outcome-independent roles

Events are grouped into seismic *sequences* by order-stable single-link space-time connected components (100 km, 30 days); the sequence, not the event, is the unit of role assignment, so events linked by this fixed rule cannot straddle roles. This rule is a deterministic grouping heuristic, not a validated catalogue of all physical mainshock–aftershock relationships; unlinked true aftershocks may therefore remain in different roles. Roles are assigned by an internally specified, outcome-independent rule (fixed assignment key, temporal cutoff year 2023, development fraction 0.2, five cross-validation folds); the rule provably never reads the label. The external holdout contains events on or after 2023. The frozen internal protocol also named a secondary leave-region-out geographic evaluation, but that branch was not executed and no geographic role manifest is claimed. The reported geographic result instead summarizes the temporal-holdout predictions by  $10^\circ$  cell (Table 7); Table 6 records this deviation. Digital fingerprints of the completed temporal role assignments are published so that third parties can verify them. The partition is exhaustive and disjoint:  $1,573 + 357 + 430 + 2 = 2,362$  events, comprising 1,573 train, 357 development, 430 external temporal holdout, and 2 temporal-straddle exclusions (Table 1, Fig. 1). Construction (train plus development) is 1,930 events across 1,530 sequences; the holdout is 430 events across 354 sequences.

## 4.3 Baseline suite and exclusions

The fixed suite is B0 no-skill (constant prevalence), B1 magnitude-only logistic regression, B2 source-only logistic regression, B3 expanded-metadata logistic regression (used only in compatible sensitivity analyses), B4 random forest

(scikit-learn; Pedregosa et al. 2011), and B5 XGBoost (Chen and Guestrin 2016). Each learned baseline is tuned *only* by the protocol-defined deterministic grouped cross-validation on the construction corpus over a fixed grid, named before evaluation, and carried unchanged to the protected holdout with frozen best parameters. A machine-readable fit log enforces a maximum of 120 model fits; the executed run used 91. Two comparators are excluded for protocol-defined, reproducible reasons recorded in the protocol-decision table (Section 5.7): B3 is not evaluated on the temporal holdout because its feature set carries origin year, a temporal proxy; and the Atkinson–Wald (2007) DYFI intensity-prediction equation (Atkinson and Wald 2007) is excluded because it cannot be applied to the source-only feature set and tectonically heterogeneous population without inventing a site-distance covariate, a per-event tectonic-regime assignment, and a probability link.

## 4.4 Metrics, calibration, uncertainty, and ranking

The primary score is the Brier score (Brier 1950); secondary scores are log-loss, AUROC, and calibration intercept/slope (Van Calster et al. 2019). For numerical stability, log-loss clips predicted probabilities symmetrically to  $[10^{-15}, 1 - 10^{-15}]$  before taking logarithms. Brier-score uncertainty is quantified by a sequence-cluster bootstrap ( $n_{\text{boot}} = 2000$ , 95% intervals) that resamples whole sequences rather than events (Field and Welsh 2007), with a minimum-valid-replicate rule. Calibration and feasibility states are *derived*, not hard-coded: a constant-prevalence predictor has no identifiable calibration slope, so the table reports that value as unavailable. Pairwise ranking uses a paired sequence-cluster bootstrap Brier difference with an explicit tie probability; the table reports whether each 95% interval includes zero, together with a tie-aware joint ordering (Section 5.5). Because the intervals use 2,000 Monte Carlo draws, near-zero endpoints are interpreted cautiously rather than as stable ranking guarantees. The protocol requested intervals for every metric; only Brier-based intervals were retained, so other metrics are descriptive point estimates (Table 6).

## 4.5 Protected evaluation and reproducibility

The official decision-bearing protected evaluation was authorized once by an automated, globally consumed token; the one-shot wrapper refuses a second authorized run. After the result was fixed, the token-free deterministic analytic core—including holdout scoring—was replayed twice for verification, and both replays reproduced the recorded result fingerprint exactly. Thus, single-use refers to result authorization and decision-bearing use, not to the total number of post-result computations. The frozen holdout predictions were also recovered bit-for-bit from the released rows and frozen best parameters (reported in Section 5.7), under a pinned scientific-Python environment (NumPy, SciPy, scikit-learn, XGBoost; Harris et al. 2020; Virtanen et al. 2020; Pedregosa et al. 2011; Chen and Guestrin 2016). All fits, exclusions, subsets, cutoffs, and pairwise comparisons are reported regardless of outcome; the only permitted omissions are the protocol-defined exclusions above. Computation is local-only under a pinned environment.

### *Reproducible artwork and generative AI assistance.*

All figures were generated with Matplotlib 3.11.1 directly from the frozen analysis assets; no manual image manipulation or AI-generated artwork was used. ChatGPT assisted language editing, formatting, code review, literature discovery, critical consistency review, and revision suggestions. It did not generate data, execute model fits, or create the reported numerical results. The authors independently verified every cited source, computation, interpretation, and assisted change; all scientific judgements and final decisions remained with the authors, who retain full responsibility for the manuscript and released artifacts.

## 5 Results

All result-bearing values below are generated from the frozen analysis outputs and checked automatically against their source records; no result cell was hand-edited.

## 5.1 Cohort and roles

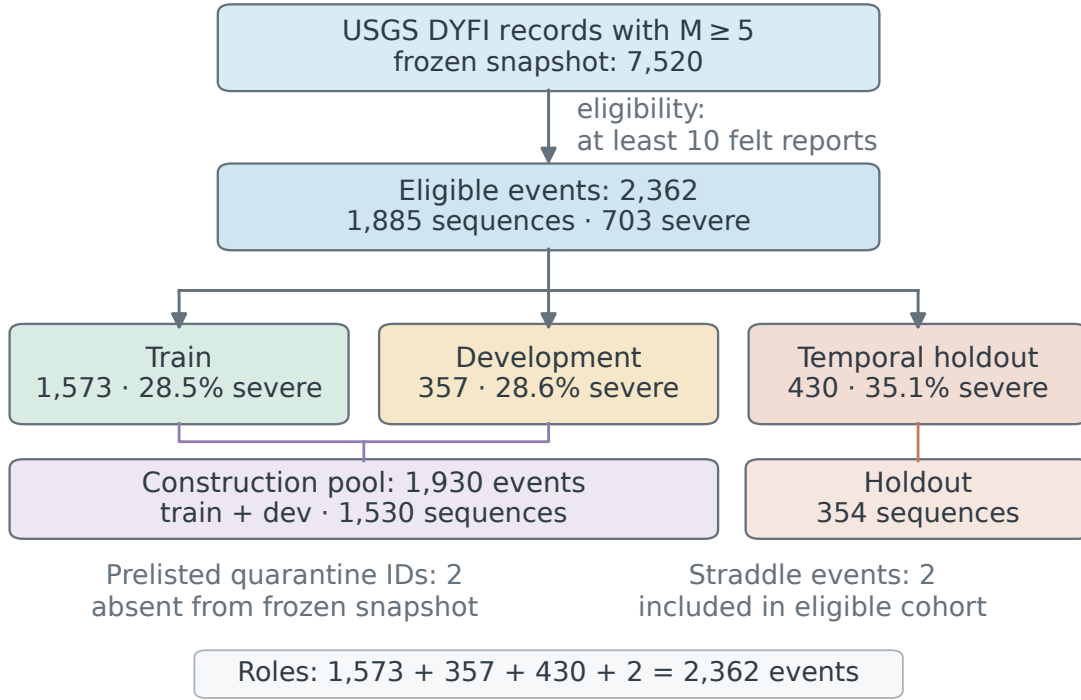
Table 1 and Fig. 1 give the cohort accounting. Of 7,520 frozen DYFI records, 2,362 events pass the ten-response eligibility filter (severe prevalence 0.298; 703 severe events), partitioned into 1,573 train, 357 development, and a 430-event temporal holdout, with two temporal-straddle exclusions. Thus 2,360 of the 2,362 eligible events receive analysis roles; they contain 702 severe events, and the 703rd severe event is one of the two straddle exclusions. Two identifiers prelisted for quarantine are absent from the frozen snapshot, so that safeguard removes no version 1 record. Construction severe prevalence is 0.285 and holdout prevalence is 0.351; the higher holdout prevalence reflects the temporal transport shift and is reported, not adjusted.

## 5.2 Difficulty and endpoint sensitivity

Table 2 and the left panel of Fig. 2 show the endpoint is not an artifact of a single threshold: as the severe cut moves from 5.5 to 6.5, holdout prevalence falls smoothly (0.458, 0.351, 0.260) and the source-only logistic-regression Brier score and AUROC move monotonically (Brier 0.240, 0.181, 0.143; AUROC 0.734, 0.786, 0.815). On the construction corpus the near-threshold fraction ( $|\text{maxCDI} - 6| < 0.5$ ) is 0.164 and the no-skill Brier score is 0.204, bounding the difficulty of the primary endpoint. This sensitivity check relabels and rescores the fixed B2 probabilities; it does not refit or recalibrate B2 at 5.5 or 6.5. The protocol-required aggregation-grid sensitivity was not executed and is disclosed as a deviation (Table 6).

## 5.3 Observed split-performance differences

The resource’s headline quantity is the optimism of naive random splitting relative to the sequence-grouped protocol, measured for a fixed leakage diagnostic—a random forest with 300 trees and unrestricted depth—under three fold-assignment schemes on the construction corpus (Table 3, Fig. 3). This diagnostic is distinct from the tuned temporal-holdout baseline B4, whose maximum depth is 3. On this  $M \geq 5$  one-row-per-event universe the point estimates are small and



**Fig. 1** Cohort construction and outcome-independent role assignment on the frozen USGS DYFI  $M \geq 5$  snapshot. The 7,520 source records yield 2,362 eligible events in 1,885 seismic sequences. Of these, 2,360 are assigned to train ( $n = 1,573$ ), development ( $n = 357$ ), or temporal holdout ( $n = 430$ ); two straddle events are excluded from role assignment

**Table 1** Cohort accounting and outcome-independent role assignment for the frozen USGS DYFI  $M \geq 5$  snapshot.

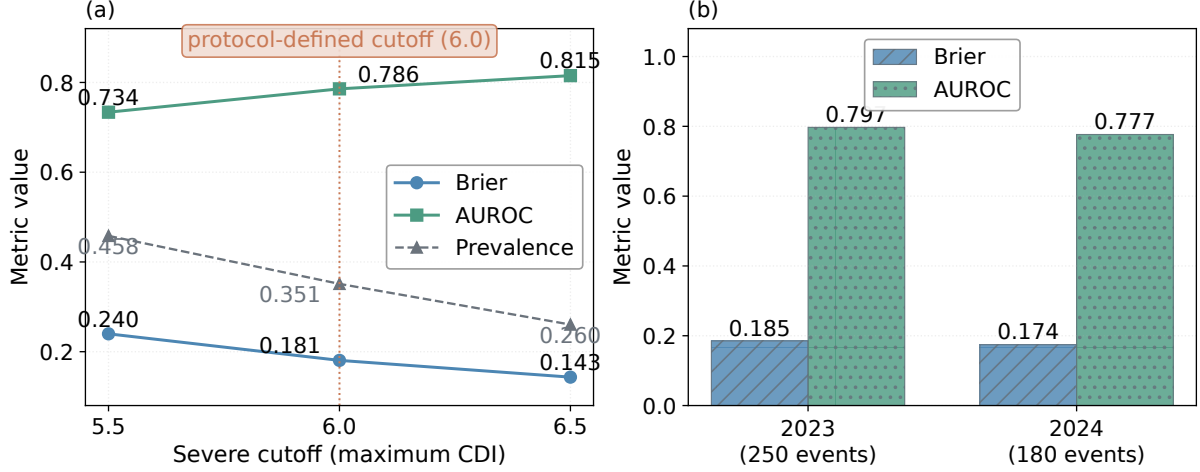
| Stage                                               | Events | Severe | Severe rate | Sequences |
|-----------------------------------------------------|--------|--------|-------------|-----------|
| USGS DYFI records with $M \geq 5$ (frozen snapshot) | 7520   | –      | –           | –         |
| Eligible (at least 10 felt reports)                 | 2362   | 703    | 0.298       | 1885      |
| Train                                               | 1573   | 449    | 0.285       | –         |
| Development                                         | 357    | 102    | 0.286       | –         |
| Construction (train+dev)                            | 1930   | 551    | 0.285       | 1530      |
| External temporal holdout                           | 430    | 151    | 0.351       | 354       |
| Temporal-straddle excluded                          | 2      | 1      | –           | 1         |

*Note.* Roles were fixed without using outcomes. Two event identifiers prelisted for quarantine are absent from the frozen snapshot, so the safeguard removes no version 1 record. Severe = 1 when an event’s maximum community decimal intensity (CDI) is at least 6. The eligible severe count reconciles as  $703 = 449$  (train) +  $102$  (development) +  $151$  (holdout) +  $1$  (straddle exclusion). Thus the 2360 role-assigned events contain 702 severe events.

slightly reversed: sequence grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than random-event splitting, calculated from unrounded scores. The sequence-grouped split therefore scores marginally *better* than the naive random split, not worse. The frozen internal protocol required uncertainty reporting for

this contrast but did not define a paired interval estimator; the executed result contains point estimates only. We disclose this unfulfilled element as a deviation and do not invent a post-hoc interval. Accordingly, we report the contrast descriptively, not as evidence of equivalence, a true zero effect, leakage, or a win. Its plausible structural context





**Fig. 2** Sensitivity analyses for the source-only logistic regression (B2). (a) Performance as the severe-intensity cutoff varies. (b) Performance within the 2023 ( $n = 250$ ) and 2024 ( $n = 180$ ) portions of the temporal holdout. The cutoff analysis rescores fixed B2 probabilities without refitting. Lower Brier score and higher AUROC are better

**Table 2** Sensitivity to the severe-intensity cutoff on the temporal holdout, evaluated with the source-only logistic regression (B2).

| Maximum CDI cutoff | Severe rate | Brier score | AUROC |
|--------------------|-------------|-------------|-------|
| 5.5                | 0.458       | 0.240       | 0.734 |
| 6.0 <sup>a</sup>   | 0.351       | 0.181       | 0.786 |
| 6.5                | 0.260       | 0.143       | 0.815 |

*Note.* <sup>a</sup> Protocol-defined primary cutoff ( $\geq 6$ ). CDI denotes community decimal intensity. The sensitivity rows relabel and rescore fixed B2 probabilities; B2 is not refit or recalibrated for the alternate cutoffs. In the construction sample, 0.164 of events lie within 0.5 CDI units of the primary cutoff, and the no-skill Brier score is 0.204. Lower Brier score and higher AUROC indicate better performance.

is that 1,885 sequences are largely singleton (1,530 in construction), leaving limited within-sequence near-duplicate structure for a random split to leak. This does not prove a small true effect. With one row per event, event grouping imposes the same grouping constraint as random-event assignment, but its deterministic fold allocation differs and therefore produces different scores. It is reported as an intermediate control. The protocol remains reusable on datasets with more multi-event sequences. The present result neither demonstrates practically important leakage from random splitting nor establishes its absence.

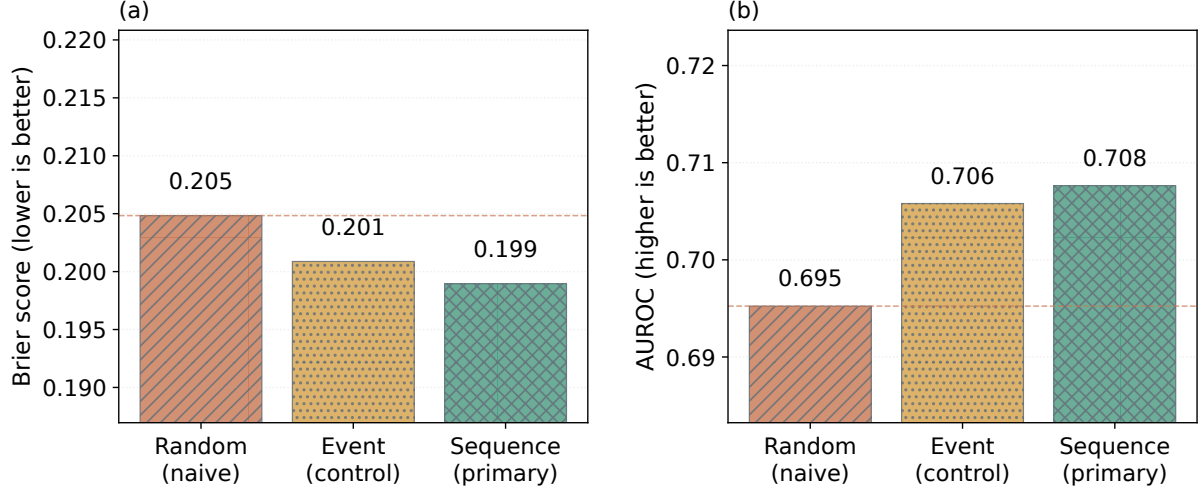
## 5.4 Baseline performance and calibration

Table 4 and Fig. 4 give per-baseline holdout performance ( $n = 430$ ) with sequence-cluster bootstrap 95% intervals. The no-skill B0 has Brier

0.232 ([0.212, 0.253]); its calibration slope cannot be estimated because it assigns the same probability to every event. The learned baselines have lower Brier (B1 0.198, B2 0.181, B4 0.187, B5 0.181) with overlapping intervals; calibration intercept/slope pairs are finite and positive. No baseline is claimed to win; the table reports proper scores, uncertainty, and derived calibration states only.

## 5.5 Pairwise ranking stability

Table 5 reports all ten pairwise Brier-score comparisons. Every learned baseline performs better than the no-skill reference B0 (all four intervals exclude zero). Among the learned models, the B2–B4 and B2–B5 comparisons have intervals that include zero. The remaining learned-model intervals exclude zero, although the B4–B5 lower endpoint (0.0003) is near the Monte Carlo boundary. In the joint bootstrap ordering, B2 has the



**Fig. 3** Five-fold performance of the fixed leakage diagnostic (random forest: 300 trees, unrestricted depth) under random-event, event-grouped, and sequence-grouped fold assignment in the construction sample. Relative to random-event splitting, sequence grouping produced (a) a Brier score 0.0059 lower and (b) an AUROC 0.0124 higher. The diagnostic is distinct from tuned temporal-holdout baseline B4. The small, reversed differences provide no evidence of performance inflation from random splitting in this sample; dashed lines mark the corresponding random-event score in each panel

**Table 3** Cross-validated performance of a fixed leakage diagnostic (random forest: 300 trees, unrestricted depth) under three fold-assignment schemes in the construction sample.

| Fold assignment            | Brier score | AUROC  | Log-loss | Folds |
|----------------------------|-------------|--------|----------|-------|
| Random-event (naive)       | 0.2048      | 0.6952 | 0.9640   | 5     |
| Event-grouped (control)    | 0.2009      | 0.7058 | 0.9223   | 5     |
| Sequence-grouped (primary) | 0.1990      | 0.7076 | 0.9709   | 5     |

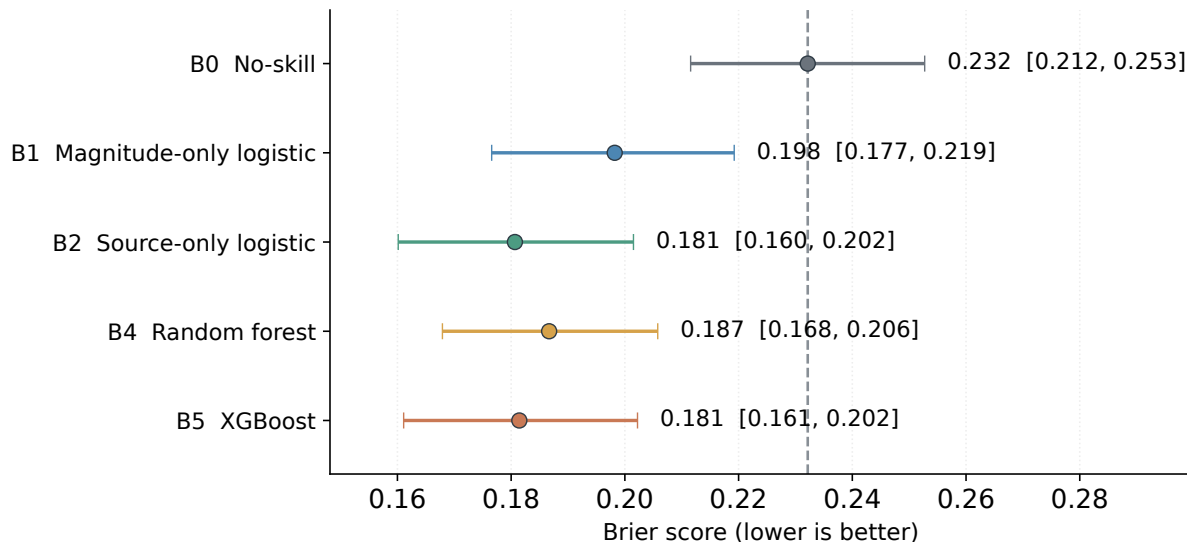
*Note.* The diagnostic is distinct from the tuned temporal-holdout baseline B4 (depth 3). Using unrounded values, sequence grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than random-event splitting. The frozen internal protocol required uncertainty but did not define the paired interval estimator; the executed report retained point estimates only. This deviation is disclosed, and no post-hoc interval or equivalence claim is added. Because there is one row per event, event grouping imposes the same grouping constraint as random-event assignment, but its deterministic fold allocation is different and can produce different scores.

highest probability of being uniquely best (0.613), followed by B5 (0.387), but the direct B2–B5 comparison remains uncertain. These results therefore do not justify naming a winning model.

The frozen internal protocol allowed an “equiv-  
alent” verdict but did not specify a positive numerical equivalence margin; its resolution floor was zero. We therefore do not invent a post-hoc margin or label any models equivalent. Instead, we report each point difference, 95% interval, and whether that interval includes zero. Table 6 records this interpretive restriction, which changes no result.

## 5.6 Temporal and geographic subgroup summaries

Table 7 and the right panel of Fig. 2 report the temporal and geographic subgroup summaries for the source-only logistic regression within the temporal holdout. The two holdout years are internally consistent (2023:  $n = 250$ , Brier 0.185, AUROC 0.797; 2024:  $n = 180$ , Brier 0.174, AUROC 0.777). The geographic subgroup summary spans 82 occupied  $10^\circ$  cells (per-cell Brier 0.002–0.825, median 0.151). These values stratify the same 430 temporal-holdout predictions; they are not a leave-region-out evaluation and support no geographic or cross-system generalization claim.



**Fig. 4** Brier scores and 95% sequence-cluster bootstrap confidence intervals for the five baselines on the temporal holdout ( $n = 430$ ). Lower values are better; the vertical dashed line marks the no-skill B0 score. B3 is omitted because it contains event year, which would reveal the temporal holdout period (Table 6)

**Table 4** Per-baseline performance on the temporal holdout ( $n = 430$ ) with sequence-cluster bootstrap 95% CIs and derived calibration states.

| Baseline                     | Brier | 95% CI         | Log-loss | AUROC | Calibration <sup>a</sup> | Accuracy |
|------------------------------|-------|----------------|----------|-------|--------------------------|----------|
| B0 (no-skill)                | 0.232 | [0.212, 0.253] | 0.658    | 0.500 | INFEASIBLE               | 0.65     |
| B1 (magnitude-only logistic) | 0.198 | [0.177, 0.219] | 0.583    | 0.719 | 0.32 / 1.02              | 0.70     |
| B2 (source-only logistic)    | 0.181 | [0.160, 0.202] | 0.536    | 0.786 | 0.64 / 1.29              | 0.73     |
| B4 (random forest)           | 0.187 | [0.168, 0.206] | 0.553    | 0.777 | 0.74 / 1.48              | 0.73     |
| B5 (XGBoost)                 | 0.181 | [0.161, 0.202] | 0.541    | 0.783 | 0.42 / 1.07              | 0.74     |

*Note.* <sup>a</sup> Intercept / slope. These values cannot be estimated for B0 because it assigns the same probability to every event. B3 includes event year and is therefore omitted from the temporal holdout (Table 6). Accuracy uses a fixed probability cutoff of 0.5 that was not selected on the holdout. Log-loss clips probabilities to  $[10^{-15}, 1 - 10^{-15}]$ . Lower Brier score and log-loss are better; higher AUROC and accuracy are better. The two lowest-Brier models cannot be distinguished in their direct paired comparison.

## 5.7 Exclusions, deviations, and reproducibility

Table 6 explains the protocol decisions most relevant to interpretation: excluded comparators and records, the unexecuted geographic transport branch, the unfulfilled split-contrast interval, and the absence of an equivalence claim. The tagged public release contains the complete SHA-256 fingerprints, software environment, and machine-readable verification details. Its public-release manifest fingerprints every distributed file. Three original execution-bookkeeping records are self-referential—they cannot contain their own

final fingerprints—and are therefore reported separately rather than treated as analysis-file mismatches.

## 6 Discussion

### *Benchmark utility.*

The resource’s utility does not rest on any model outcome. It rests on four reusable properties, each demonstrated here: an executable leakage audit that rejects seeded contamination and accepts only the approved feature set; internally specified outcome-independent roles that a third party can reproduce; calibrated, uncertainty-bearing baselines with a genuine no-skill comparator; and exact reconstruction anchored by recorded digital

**Table 5** Pairwise comparison of baseline Brier scores on the temporal holdout using a paired sequence-cluster bootstrap.

| Model A | Model B | A – B   | 95% CI            | $P(A \text{ better})$ | CI relation |
|---------|---------|---------|-------------------|-----------------------|-------------|
| B0      | B1      | 0.0339  | [0.0171, 0.0511]  | 0.000                 | excludes 0  |
| B0      | B2      | 0.0515  | [0.0342, 0.0681]  | 0.000                 | excludes 0  |
| B0      | B4      | 0.0455  | [0.0307, 0.0598]  | 0.000                 | excludes 0  |
| B0      | B5      | 0.0507  | [0.0319, 0.0682]  | 0.000                 | excludes 0  |
| B1      | B2      | 0.0176  | [0.0101, 0.0249]  | 0.000                 | excludes 0  |
| B1      | B4      | 0.0115  | [0.0023, 0.0205]  | 0.009                 | excludes 0  |
| B1      | B5      | 0.0168  | [0.0055, 0.0273]  | 0.001                 | excludes 0  |
| B2      | B4      | -0.0060 | [-0.0118, 0.0002] | 0.970                 | includes 0  |
| B2      | B5      | -0.0008 | [-0.0078, 0.0068] | 0.613                 | includes 0  |
| B4      | B5      | 0.0052  | [0.0003, 0.0099]  | 0.017                 | excludes 0  |

*Note.* Lower Brier score is better, so a negative A – B difference favors model A. Intervals use 2,000 Monte Carlo bootstrap draws; near-zero endpoints are interpreted cautiously and not as stable ranking guarantees. Because no numerical equivalence margin was specified before analysis, intervals including zero are not labelled equivalent. B0 is the no-skill reference; B1 is magnitude-only logistic regression; B2 is source-only logistic regression; B4 is random forest; and B5 is XGBoost.

**Table 6** Excluded analyses and disclosed deviations affecting interpretation.

| Decision                          | Treatment                     | Rationale                                                                                                           |
|-----------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------|
| B3 expanded-metadata model        | Omitted from temporal holdout | Includes event year, which reveals the temporal evaluation period.                                                  |
| Published DYFI intensity equation | Omitted as comparator         | Needs unavailable site distance, tectonic-region inputs, and probability conversion.                                |
| Two prelisted quarantine IDs      | No effect in version 1        | Both are absent from the frozen snapshot; no released record is excluded.                                           |
| Geographic transport evaluation   | Not executed                  | Only temporal-holdout geographic subgroups are reported; no leave-region-out or generalization claim.               |
| Aggregation-grid sensitivity      | Not executed                  | Required but absent; only threshold sensitivity of fixed predictions is reported.                                   |
| Uncertainty coverage              | Partial deviation             | The protocol requested intervals for every metric and the split contrast; only Brier-based intervals were retained. |
| Equivalence interpretation        | Descriptive only              | No numerical margin was specified; intervals are reported without equivalence labels.                               |

*Note.* These decisions were recorded before, or disclosed with, the analysis. No result, subset, cutoff, or comparison was omitted because of its observed outcome.

fingerprints and a single-use authorization safeguard. A later study can evaluate a new method against this fixed contract without reconstructing a contaminated split, which is the practical benefit the field currently lacks (Patelli et al. 2026; Mori and Naso 2025; Roberts et al. 2017).

### *Interpreting the split contrast.*

The observed point estimate is small and slightly reversed. In these folds the naive random split did not inflate the diagnostic score, but the missing interval prevents an equivalence, zero-effect, or “little practical effect” inference. The protocol

remains useful for datasets containing more multi-event sequences; this result neither demonstrates leakage nor supports a win.

### *No winning model.*

Every learned baseline separates from the no-skill reference. However, the direct B2–B5 comparison between the two lowest-Brier models is unresolved, as is B2–B4, so the results do not establish a decisive winner. Consistent with the resource’s scope, we claim no winning model and no operational, real-time, lead-time, cross-system, or causal benefit.

**Table 7** Temporal and geographic subgroup summaries within the temporal holdout for the source-only logistic regression (B2).

| Subset                                        | Events | Brier score  | AUROC |
|-----------------------------------------------|--------|--------------|-------|
| Temporal 2023                                 | 250    | 0.185        | 0.797 |
| Temporal 2024                                 | 180    | 0.174        | 0.777 |
| Geographic cells within temporal holdout (82) | 430    | 0.151 (med.) | –     |

*Note.* The geographic row partitions the same 430 temporal-holdout predictions into 82 occupied  $10^\circ$  cells. It is a descriptive subgroup summary, not a leave-region-out or external geographic evaluation. The entry is the median cell-level Brier score; cell values range from 0.002 to 0.825 and are included in the public release.

## 7 Limitations and threats to validity

### *Outcome and catalogue limitations.*

The endpoint is magnitude-dominated, CDI remains sensitive to responder coverage, and magnitude and depth are frozen preferred catalogue values rather than guaranteed first-alert values. The ten-response rule mitigates ascertainment; response count is used only for eligibility. B1 provides the simple magnitude comparator.

### *Unexecuted sensitivity and transport branches.*

The geographic values are temporal-holdout subgroups, not external transport evidence. The protocol-defined geographic evaluation and aggregation-grid sensitivity were not executed, and no alternative to the fixed 100-km/30-day sequence definition was tested. The rule guarantees co-assignment only for events it links; it does not guarantee identification of every physical aftershock relationship. Consequently, no geographic, aggregation, sequence-definition, or complete aftershock-isolation robustness claim is made.

### *Inference and held-out semantics.*

The protocol-required split-contrast uncertainty interval was not retained, so that result remains descriptive with no equivalence or zero-effect inference. Non-Brier scores remain descriptive point estimates. Pairwise intervals use 2,000 Monte Carlo draws and near-zero endpoints are interpreted cautiously. The temporal holdout was excluded from model development, but aggregate feasibility was previously known; it is neither “untouched” nor confirmatory.

### *Public release.*

Version 1 of the data, code, frozen results, and reproduction instructions is available in a tagged public repository at <https://github.com/akssha74/dyfi-benchmark-reliability>. The release retains complete digital fingerprints so that later revisions remain distinguishable from the version evaluated here.

## 8 Conclusion

We release a versioned, digitally fingerprinted, single-source USGS DYFI benchmark for retrospective severe felt-intensity prediction from outcome-blind source metadata. The benchmark provides a reusable, leakage-audited evaluation contract, reference baselines, and verification tools. On the frozen snapshot, the observed split difference is small and slightly reversed, every learned baseline separates from the no-skill reference, and comparisons among the lowest-Brier models do not establish a winner. Every reported number reproduces exactly from the frozen analysis files. The resource lets subsequent macroseismic-intensity work evaluate against a fixed leakage-audited protocol, and remains reusable on denser-sequence datasets containing more multi-event sequences.

## Statements and Declarations

- **Funding.** No specific funding was received.
- **Competing interests.** The authors have no relevant financial or non-financial interests to disclose.
- **Ethics approval.** Not applicable. The study analyzes public, aggregate event-level USGS products rather than individual questionnaire



responses; it involved no interaction with participants, identifiable data, human-subject collection, animals, or biological material.

- **Consent to participate.** Not applicable; no participants were recruited.
- **Consent for publication.** Not applicable.
- **Data availability.** Version 1 data, results, manifests, and instructions are available under CC0 1.0 at <https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.0>. 25. Source data are USGS DYFI and ANSS ComCat (public domain; [10.5066/F7J101C8](https://doi.org/10.5066/F7J101C8) and [10.5066/F7MS3QZH](https://doi.org/10.5066/F7MS3QZH)) and are reconstructable via the released fetch-and-verify source-locator manifest. No respondent-level personally identifying information is redistributed.
- **Code availability.** MIT-licensed resource code and a standard-library printed-value verifier are provided in the same tagged release. Full prediction reconstruction and model tests use the separately pinned scientific-Python environment.
- **Author contributions.** A. Sharma designed the benchmark protocol and analysis and drafted the manuscript. L. Prasad supervised and revised the work. Both authors approved the final manuscript.

## References

- Atkinson GM, Wald DJ (2007) “Did you feel it?” intensity data: A surprisingly good measure of earthquake ground motion. *Seismological Research Letters* 78(3):362–368. <https://doi.org/10.1785/gssrl.78.3.362>
- Brier GW (1950) Verification of forecasts expressed in terms of probability. *Monthly Weather Review* 78(1):1–3. [https://doi.org/10.1175/1520-0493\(1950\)078<0001:VOFEIT>2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2)
- Chen T, Guestrin C (2016) XGBoost: A scalable tree boosting system. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp 785–794, <https://doi.org/10.1145/2939672.2939785>
- Ertuncay D, de Lorenzo A, Costa G (2025) Deep learning based earthquake and vehicle detection algorithm. *Journal of Seismology* 29(1):269–281. <https://doi.org/10.1007/s10950-024-10267-8>, first published online in 2024
- Fathian Sabet A, Abrahamson N, Parker GA, et al (2026) Macroseismic Intensity Dataset in Western North America. U.S. Geological Survey data release, <https://doi.org/10.5066/P143CZ3R>, URL <https://doi.org/10.5066/P143CZ3R>
- Field CA, Welsh AH (2007) Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 69(3):369–390. <https://doi.org/10.1111/j.1467-9868.2007.00593.x>
- Gebru T, Morgenstern J, Vecchione B, et al (2021) Datasheets for datasets. *Communications of the ACM* 64(12):86–92. <https://doi.org/10.1145/3458723>
- Harris CR, Millman KJ, van der Walt SJ, et al (2020) Array programming with NumPy. *Nature* 585:357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- Karthik Kumar P, Vismay, Apoorva, et al (2025) Machine learning based DYFI intensity prediction for seismic events in India. In: *2025 9th International Conference on Electronics, Communication and Aerospace Technology (ICECA)*. IEEE, <https://doi.org/10.1109/iceca66444.2025.11382872>
- Kaufman S, Rosset S, Perlich C, et al (2012) Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data* 6(4):15:1–15:21. <https://doi.org/10.1145/2382577.2382579>
- Kouskouna V, Ganas A, Kleanthi M, et al (2021) Evaluation of macroseismic intensity, strong ground motion pattern and fault model of the 19 july 2019 mw5.1 earthquake west of athens. *Journal of Seismology* 25(3):747–769. <https://doi.org/10.1007/s10950-021-09990-3>

- Li L, Li D, Ou Z, et al (2025) LLMs as world models: Data-driven and human-centered pre-event simulation for disaster impact assessment. In: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, Suzhou, China, pp 3078–3096, <https://doi.org/10.18653/v1/2025.emnlp-main.153>, released DYFI-grounded MMI benchmark with a location-identifier leakage test
- Lilienkamp H, Bossu R, Cotton F, et al (2023) Utilization of crowdsourced felt reports to distinguish high-impact from low-impact earthquakes globally within minutes of an event. *The Seismic Record* 3(1):29–36. <https://doi.org/10.1785/0320220039>
- Mori F, Naso G (2025) An explainable machine learning model for predicting macroseismic intensity for emergency management. *Remote Sensing* 17(10):1754. <https://doi.org/10.3390/rs17101754>
- Neefs B, Van Noten K, Vanneste K, et al (2025) The belgian traditional macroseismic (BTM) database of the twentieth century. *Journal of Seismology* 29(5):1005–1036. <https://doi.org/10.1007/s10950-024-10266-9>, first published online in 2024
- Oliveti I, Faenza L, Michelini A (2022) INGe: Intensity-ground motion data set for Italy. *Annals of Geophysics* 65(1):DM102. <https://doi.org/10.4401/ag-8709>
- Özüpak Y (2026) Explainable and statistically validated machine learning for earthquake alert-level classification using seismic intensity features. *Journal of Seismology* 30(5). <https://doi.org/10.1007/s10950-026-10444-x>
- Patelli L, Cameletti M, De Rubeis V, et al (2026) Machine learning for prompt estimation of macroseismic intensity from seismometric data in Italy. *Scientific Reports* 16:7265. <https://doi.org/10.1038/s41598-026-35740-x>
- Pedregosa F, Varoquaux G, Gramfort A, et al (2011) Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* 12:2825–2830
- Quitoriano V, Wald DJ (2020) USGS “Did you feel it?”—science and lessons from 20 years of citizen science-based macroseismology. *Frontiers in Earth Science* 8:120. <https://doi.org/10.3389/feart.2020.00120>
- Roberts DR, Bahn V, Ciuti S, et al (2017) Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40(8):913–929. <https://doi.org/10.1111/ecog.02881>
- Rösler B, Stein S, Hough SE (2022) On the documentation, independence, and stability of widely used seismological data products. *Frontiers in Earth Science* 10:988098. <https://doi.org/10.3389/feart.2022.988098>
- Saraò A, Tamaro A, Sandron D, et al (2023) On the crowdsourcing of macroseismic data to characterize geological settings. *International Journal of Disaster Risk Reduction* 96:103934. <https://doi.org/10.1016/j.ijdrr.2023.103934>
- Sira C, Schirr L, Bertrand V, et al (2025) MACROSISDATA: Safeguarding and promoting the heritage of french macroseismic surveys carried out between 1921 and 1996. *Journal of Seismology* 29(5):1145–1160. <https://doi.org/10.1007/s10950-025-10322-y>
- Stockman S, Lawson DJ, Werner MJ (2026) EarthquakeNPP: A benchmark for earthquake forecasting with neural point processes. *Transactions on Machine Learning Research* URL <https://openreview.net/forum?id=dIcNAg6ZuZ>, article 6105
- U.S. Geological Survey, Earthquake Hazards Program (2017) Advanced national seismic system (ANSS) comprehensive catalog of earthquake events and products. U.S. Geological Survey, <https://doi.org/10.5066/F7MS3QZH>
- Van Calster B, McLernon DJ, van Smeden M, et al (2019) Calibration: The Achilles heel of predictive analytics. *BMC Medicine* 17:230. <https://doi.org/10.1186/s12916-019-1466-7>

- Virtanen P, Gommers R, Oliphant TE, et al (2020) SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods* 17:261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- Wald DJ, Dewey JW (2005) Did You Feel It? U.S. Geological Survey data release, <https://doi.org/10.5066/F7J101C8>
- Wald DJ, Quitoriano V, Dengler LA, et al (1999) Utilization of the Internet for rapid community intensity maps. *Seismological Research Letters* 70(6):680–697. <https://doi.org/10.1785/gssrl.70.6.680>
- Wald DJ, Quitoriano V, Worden CB, et al (2012) USGS “Did you feel it?” internet-based macroseismic intensity maps. *Annals of Geophysics* 54(6):688–707. <https://doi.org/10.4401/ag-5354>
- Wilkinson MD, Dumontier M, Aalbersberg IJ, et al (2016) The FAIR guiding principles for scientific data management and stewardship. *Scientific Data* 3:160018. <https://doi.org/10.1038/sdata.2016.18>